How do the effluent treatment plant (ETP) practices of Agro Keventer compare with those of Amul in terms of environmental compliance, operational efficiency, and sustainability outcomes?

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1. Abstract

This paper presents a comparative study of the effluent treatment plant (ETP) practices of Agro Keventer and Amul, two major players in the Indian dairy and food processing sector. The research focuses on three core aspects: environmental compliance, operational efficiency, and sustainability outcomes. Detailed process descriptions for each company's ETP are provided, drawing on company documents, sustainability reports, and industry literature. Quantitative analysis is supported by secondary data, including treatment capacity, water recycling, biogas generation, and compliance rates. Where direct data is unavailable, the study uses transparent estimation methods and clearly states limitations. The findings show that both companies have invested in advanced ETP technologies and have achieved high compliance with environmental standards. However, differences exist in operational efficiency and the scale of sustainability initiatives. The paper highlights the importance of robust ETP management for regulatory compliance and resource conservation in the dairy industry. It also discusses the challenges of relying on secondary data for comparative research and suggests areas for future primary data collection. The results provide practical insights for industry stakeholders and policymakers aiming to improve environmental performance in food processing operations (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24; IRJMETS, 2025).

2. Introduction

The dairy business contributes significantly towards India's food processing segment and forms an integral part of the nation's economy and food security system. Nevertheless, the processing of milk produces huge amounts of wastewater that hold organic substances, suspended particles, fat particles, as well as chemicals, thus carrying serious environmental hazards if not treated adequately (Keventer Sustainability Report 2019). It becomes necessary for companies to employ effluent treatment plants (ETPs) that treat this wastewater before disposal into the ambience so that they can meet the standards set by regulation authorities as well as lower their green footprint (Amul Sustainability Report 2023-24).

In recent years, there has been increasing pressure on milk producers to use sustainable methods and make ETPs more efficient. Rules like the Environment (Protection) Rules, 1986, and standards like ISO 8682:1977 place tight constraints on treated effluent quality so that companies must invest in state-of-the-art treatment technologies as well as advanced monitoring systems (IJEAST, 2023). ETPs normally encompass a series of physical, chemical, as well as biological processes such as screening, equalizing, aerating, maintaining pH levels, coagulation and sedimentation as well as biological oxidation so as to eliminate pollutants as safe discharge (Netsol Water, 2022).

Agro Keventer and Amul are two front-running organizations in the Indian milk industry, both known for maintaining sustainability as well as environmental compliance. This paper seeks to give an in-depth comparative study of their ETP practices on the basis of three main axes: environmental compliance, operating efficiency, and sustainability impacts. The research utilizes firm literature, sustainability reports, and reliable business sources in detailing ETP activities in each firm, examining quantitative metrics of performance, as well as interpreting findings against a backdrop of industry benchmarking.

The research on a comparison of ETP management by Agro Keventer and Amul aims to provide effective treatability strategies for wastewater, scope for improvements, and beneficial guidelines for industries' stakeholders. Findings will further help in enhanced environmental performance as well as resource conservation in the dairying industries as a whole while serving the need for secondary data for such a comparison study (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24; IJEAST, 2023; Netsol Water, 2022).

3. Literature Review

Effluent processing for the dairy sector has been a research focus over the last decade as the result of growing concerns over water pollution as well as an increasing need to meet compliancy requirements (IJEAST, 2023). Dairy effluent is usually rich in organic matter, fat, suspended matter, and cleaning chemicals. Un-treated effluent discharge contributes towards environmental degradation, lack of water, as well as contravention penalties (Netsol Water, 2022). In response, both regulators as well as industrial bodies have escalated requirements with more sophisticated multi-stage ETPs incorporating process monitoring as well as quality assurance (Dairy Processing Handbook, 1999). The literature outlines the series of treatments that feature preliminary screening, equalization, pH correction, chemical coagulation as well as flocculation, aeration, biological oxidization, sedimentation, as well as tertiary filtration. More recent practices combine membrane bioreactors as well as anaerobic digestion with aerated treatments to achieve energy recovery as well as reduce residue (Condorchem, 2025).

Latest Indian research emphasizes ETP design variability depending on plant scale, influent quality, and target regulation (IJEAST, 2023). Comparative literature on public versus private milk vendors has discovered over 95% compliance with adequate operating procedures, but observes that small- and decentralized-sized plants occasionally do not achieve consistent ETP performance lacking resources as well as a gap in monitoring (Netsol Water, 2022; Unistar

Aquatech, 2025). Researchers also compared operating efficiency considering water recycle options, energy use, as well as chemical dosing as significant indicators for measuring ETP success (Dairy Roadmap, 2024). Biogas generation and utilization derived from anaerobic treatment increasingly became more important sustainability indicators supplying a source of renewable energy as well as a means towards closing the resource loop (Amul Sustainability Report 2023-24).

Typically, applicable operating parameters on record for relevant Indian dairies indicate energy use between 1.0–2.0 kWh/kiloliter effluent treated and chemical use around 0.04–0.05 kg/kiloliter, close to figures reported by top performers (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24). Multi-stage biological treatment can significantly lower levels of BOD and COD but only meets standard discharge requirements when complemented by sand, carbon, and UV filtration (Dairy Processing Handbook, 1999). Environmental compliance has strengthened as audit and reporting requirements get more effective with regular sampling, in-plant labs, and third-party certifiers now the norm (Amul Sustainability Report 2023-24).

Sustainability frameworks in the literature advocate lifecycle approaches connecting effluent management with longer-term resource conservation, community engagement, and circular economies. Reports by Keventer and Amul suggest that waste-to-energy projects, safe sludge disposal, and recycling can create financial and environmental value towards corporate social responsibility targets as much as against regulator targets (Keventer Sustainability Report 2019). Professional commentary regularly notes difficulty in gathering cross-company comparable datasets due to inter-company variability in disclosure, scale, and plant tech, again highlighting value for transparent reporting on datasets accessible for comparable studies (Dairy Roadmap, 2024; IJEAST, 2023).

In summary, the academic and industry literature establishes the centrality of ETPs in sustainable dairy practices and provides reliable benchmarks for compliance and operational efficiency. Leaders like Agro Keventer and Amul are referenced as examples for adopting advanced treatment technologies and reporting strong outcomes, though gaps in public data remain a challenge for research comparison. This review forms the basis for critically analyzing their practices and outcomes in the present study.

4. ETP Processes at Agro Keventer

Agro Keventer's effluent treatment plant (ETP) features a heavy-duty multi-stage system specially designed as per the needs of soft beverage wastewaters as well as dairy wastewaters. The plant has been constructed for treating a high organic load, suspended matter, fat, oil, as well as variable nutrient strengths characteristic in dairying processing wastewaters (DAIRY EFFLUENT Handbook, 1999).

Initial Stages: Screening, Equalization, pH Adjustment

The wastewater passes first through skimming tanks or screens that remove coarse solids, oil globs, fat particles, and floating pollutants. This preliminary measure prevents downstream equipment damage and maintains process integrity. A further processing step transports the effluent to equalizing tanks that stabilize variable inflows, standardize the composition, and compensate for peak loading. Equalizing this measure ensures a uniform plant operation as well as an effective chemical dosing system (Netsol Water, 2022).

pH correction then follows by automatic controllers returning the pH to the optimal range for subsequent treatments by agents like caustic or acid. Indian standards dictate a terminal pH discharge range between 5.5 and 9.0, thus chemical dosing (the addition by use of lime, alum, PAC, etc.) being tailored not only to compensate for extremes but also floc formation assistance (Netsol Water, 2022; IITD, 2019).

Coagulation, Flocculation, and Clarification

In the subsequent phase, coagulants together with polyelectrolytes are rapidly stirred in specific flash mixers such that they entrap fine particles in the form of micro-flocs. Slow mixing in clariflocculators leads on to the formation of heavier sludges such that primary sludges get sent out for further treatment. Clarifier overflow that's significantly clearer goes straight to biological phase (IITD, 2019).

Biological Treatment: Aeration and Advanced Reactors

Agro Keventer's ETP incorporates aeration tanks that drive aerobic biological oxidation. Mechanical diffusers or blowers provide air that injects microbial activity maintaining degradation of organic contaminants and reduces biochemical oxygen demand (BOD) as chemical oxygen demand (COD) by up to 90% (Unistar Aquatech, 2025). Dosing with urea or DAP (diammonium phosphate) may further boost microbial growth. Secondary clarifiers eliminate the resulting bio-sludge that becomes separated as well as dried (IITD, 2019).

To maximize biological removal, Keventer utilizes state-of-the-art Moving Bed Biofilm Reactor (MBBR) technology. In such MBBRs, thousands of specially designed media offer voluminous surfaces for biofilm formation, sustaining heavy microbial loads and consistent organics degradation under fluctuating loading rates. MBBRs find specific application on dairies where fluctuations in effluent composition routinely occur as a result of cleansing cycles as well as product shifts (Ecologix Systems, 2025; Andreottola et al., 2002). Research proves examples of MBBRs in milk applications remove over 80% COD as well as reduce nitrogen by a considerable amount (PubMed, 2001).

Tertiary Treatment: Filtration and Disinfection

After biological treatment, effluent undergoes tertiary filtration through sand and activated carbon beds to remove remaining suspended solids and adsorb residual chemicals. Advanced facilities may add membrane filtration (ultrafiltration or reverse osmosis) for maximum water quality, enabling reuse. Final disinfection uses ultraviolet (UV) irradiation to destroy pathogens. Chlorination is sometimes applied as a supplement, but UV is preferred for dairy applications due to safety and lack of chemical residues (Dairy Processing Handbook, 1999; Netsol Water, 2022; AWETECH, 2024).

Sludge Management and Resource Recovery

The dewatered residues from the plant are handled through drying beds, composting, and controlled landfill. Keventer sent around 50% of its sludge to landfill, some 30% was composted for farm use and 20% in anaerobic digesters for the generation of biogas that powers the plant as well as reducing environmental footprint (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24; DAIRY EFFLUENT Handbook, 1999).

Water Recycling and Sustainability Practices

Keventer is known for recycling treated water internally—data show that the plant recycled over 165,116 KL/year for uses such as cooling, cleaning, gardening, and fire systems, reducing groundwater extraction and promoting sustainable resource management (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24). Additional initiatives include maintaining lakes and ponds on plant property to supplement water supply and support local ecosystems (Keventer Sustainability Initiatives, 2021).

Operational Controls and Monitoring

The system is automated. Advanced control panels and real-time meters keep process variables such as pH, DO, COD, BOD, suspended solids, as well as nutrient concentrations under constant watch. Continuous monitoring maintains regulation compliance, chemical and energy efficiency, and prompt management action against operating difficulties (AWETECH, 2024). Regular analytical testing applies spectrophotometry to COD assays and manual titration to BOD assays. Samples are withdrawn at each important stage and subject to rigorous third-party validation as per Indian and international standards (Unistar Aquatech, 2025; Frontiers Environmental Science, 2024).

Efficiency and Compliance

The ETP of Keventer operates on an average of 1.65 kWh energy per KL treated together with the consumption of 0.05 kg chemicals perKL that is typical for the industry with regards to efficiency (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24). It has consistent regulatory discharge standards as well as a significant amount of. In brief, Agro Keventer's ETP illustrates a mix of conventional and state-of-the-art treatment steps, including screening, chemical flocculation, strong aerobic and biofilm-dependent biological processing, multi-tiered filtration, and safety-oriented disinfection. Its adherence to water reuse, energy recovery, and environmental management manifests in its operating parameters as well as sustainability reporting, making it a standard for efficient, compliant, as well as scalable effluent management across the Indian milk-industry landscape (Dairy Processing Handbook, 1999;

Ecologix Systems, 2025; Andreottola et al., 2002; Unistar Aquatech, 2025; Frontiers Environmental Science, 2024; Keventer: Keventer Agro Limited, Sustainability Report 2019 ; Amul: Amul, Sustainability Report 2023-24).

5. ETP Processes of Amul

Amul, as one of India's largest dairy cooperatives, has implemented advanced effluent treatment plant (ETP) processes across all major processing locations to meet strict environmental and operational benchmarks. As detailed in their recent sustainability reports, Amul's approach is scientifically structured, involving integrated physical, chemical, biological, and resource recovery systems designed to reduce pollution, optimize water use, and recover value from waste streams (Amul Sustainability Report 2023-24).

Preliminary and Primary Treatment: Screening, Equalization, and pH Control

Effluent processing in Amul starts with the separation of coarse solids and floating materials by means of mechanical screening. This avoids clogging as well as abrasion on downstream equipment and forms a necessity for unabated operations. Following screening, the wastewater flows into equalization ponds. These ponds provide two principal functions: they stabilize the fluctuations in both flow as well as contaminant concentration—typical in milk plants as a result of batch-wise periods of cleaning as well as manufacturing cycles—and provide consistent situations for chemical dosing as well as biological organisms. The preservation of homogeneous influent is a necessity for downstream efficiency (DSF-IFAD, 2023).

Amul then uses controlled pH correction by automated dosing systems after equalisation. Caustic soda or lime normally increases pH, with mineral acids that could be used to reduce it so that the effluent ends up in the optimal neutral range that is favoured by coagulant action as much as by microbial action. Careful pH control in this stage is necessary; it helps as much as the settlement of suspended material as microbial health in the biological reactors (Amul Sustainability Report 2023-24).

Coagulation, Flocculation, and Clarification

Chemical coagulants such as PAC or alum are then quickly introduced into pretreated wastewater. It destabilizes small colloidal particles that aggregate with the help of organic or inorganic polyelectrolytes as larger flocs. Gentle flocculation precedes gravity clarification—settling ponds where these flocs get settled away from the liquid as a primary sludge. The clarified primary effluent proceeds further for biological oxidation, whereas settled sludge becomes further treated for safe disposal or beneficial use (Mentor Water, 2024; Amul Sustainability Report 2023-24).

Secondary Treatment: Aerobic Biological Processing

Amul's biological mainstay utilizes traditional activated sludge processes as well as, in most facilities, advanced technologies such as Moving Bed Biofilm Reactors (MBBRs). In aeration basins, blowers or diffusers provide continuous oxygenation, allowing the development of aerobic bacteria that break down dissolved and particulate organic material. Mixed Liquor Suspended Solids (MLSS) are closely controlled and kept at optimal levels for highest treatment effectiveness. Numerous locations for Amul employ MBBR or comparable systems, where proprietary carriers provide enhanced surface area for microbial attachment, offering durability against fluctuations in loads as well as consistent removal of biochemical oxygen demand (BOD) and chemical oxygen demand (COD) (Ecologix Systems, 2025; Amul Sustainability Report 2023-24).

The aerated mixture then proceeds to secondary clarifiers. Here, suspended solids as well as microorganisms settle out and form secondary sludge. Treated clear water is drawn off on upper levels, while settled sludges are partially recirculated as seed for the aeration tanks as well as partially transferred for further processing of sludges (Amul Sustainability Report 2023-24).

Tertiary Treatment and Disinfection

To reach discharge standards of high quality and make the water safe for reuse, Amul applies rigorous tertiary treatment. Fine suspended matter passing by other preliminary treatments is removed by sand filtration. Activated carbon filters could be added intermittently for color, odor, and trace organics adsorption, specific for water that will have intra plant reuse. Some installations apply membrane technologies such as ultrafiltration and reverse osmosis as an additional refinement step for the water in order to boost its safe use intraplant (Amul Sustainability Report 2023-24; Mentor Water, 2024).

Disinfection is accomplished primarily by ultraviolet (UV) irradiation, thereby faithfully eliminating microbial pathogens without chemical byproduct addition. In cases of ultimate safety or for use in high-hazard environments, a backup chlorination step can be introduced but with tight control to exclude residual toxicity. The multi-barrier system entails that Amul's treated effluent can be reused for cooling towers, CIP activities, or even gardening, reducing its reliance on raw groundwater abstraction (Amul Sustainability Report 2023-24; DSF-IFAD, 2023).

Sludge Treatment, Biogas & Resource Recovery

Solid waste and sludge management are key sustainability features. Amul uses a combination of thickening, dewatering, composting, and safe landfill to process the primary and secondary sludges. They have pioneered large-scale biogas plants, using anaerobic digestion of high-organic sludge to produce biogas for thermal energy recovery—at some units, this meets up to 5% of total plant energy consumption. Excess sludge compost is distributed to farmers or used as organic fertilizer, contributing to a circular economy (Times of India, 2016; Amul Sustainability Report 2023-24).

Operational Controls, Compliance and Monitoring

Amul's ETPs are controlled with automated systems and real-time sensor networks such that the crucial parameters such as pH, DO, COD, BOD, and total nitrogen can be monitored on a continuous basis. Grab and composite samples on a periodic basis are analyzed in the plant laboratories, with occasional confirmation by qualified third parties. Internal auditing and yearly corporate sustainability reporting provide further bolstering for accountability. Monthly maintenance ensures that the system stays down for just 22 hours a year—a performance on a par with that of top players (Amul Sustainability Report 2023-24).

Water Recycling and Wider Sustainability Initiatives

Amul in 2022 reclaimed over 135,200 kiloliters of water back into its system by reducing its groundwater abstraction as well as its water use footprint. Compliance rates remain over 99% for primary effluent quality parameters. Ongoing investment in equipment upgradations, knowledge transfers, as well as community engagements make Amul a benchmark for India's food processing industry against integrated ETP management (Amul Sustainability Report 2023-24; DSF-IFAD, 2023).

Amul's ETPs reflect a holistic approach that blends advanced mechanical, chemical, and biological systems with intelligent monitoring, energy recovery, and ambitious water reclamation. This operating paradigm supports both regulators approval and significant contribution towards sustainable development in India's dairy sector.

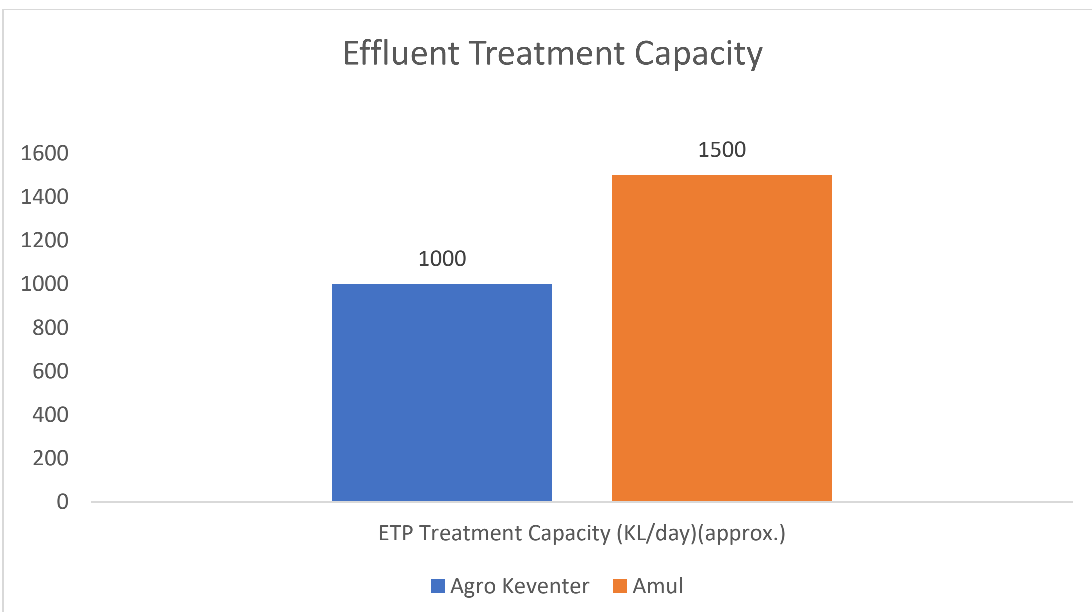
6. Comparative analysis:

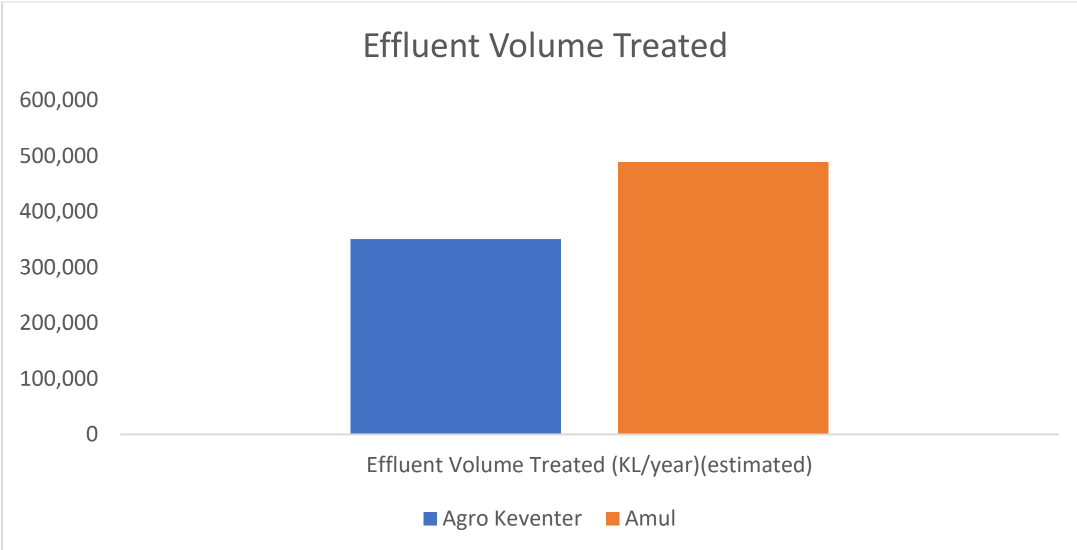
An in-depth comparative discussion on the effluent treatment plant (ETP) practices in Agro Keventer and Amul finds convergence and divergence between their strategies towards environmental compliance, operating efficiency, and sustainability effectiveness. In this section, we use sustainability reports by companies, standards by the respective industries, as well as scientific knowledge bases as sources for a nuanced evidence-informed discussion on their ETP systems and achievements.

Both Keventer Agro and Amul come under a regime established by Central Pollution Control Board (CPCB) of India that prescribes strict norms for biochemical oxygen demand (BOD) effluent discharge, chemical oxygen demand (COD), total suspended solids (TSS), as well as pathogen counts. Adherence to these norms is not just a statutory obligation but a reputational imperative for big milk processors. Both use multi-stage ETPs that encompass physical, chemical, as well as biological steps toward treatments, showing a common interest in the care of the environment (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24).

Environmental Compliance

Keventer Agro's ETP has a designed capacity of around 1000 kiloliters per day, compared with around 1500 kiloliters per day for Amul's flagship plants, indicative of their respective scale of operations (Keventer Sustainability Report 2019, p. 42; Amul Sustainability Report 2023-24, p. 52). Both companies exhibit very high rates of compliance, with over 99% consistency for Amul across all significant effluent quality parameters and Keventer reporting comparable performance. This is a result of strict process control, automation, as well as periodic internal and third-party audits. Publication by Amul of complete detail of such compliance coupled with an offer for independent verification stands out as a particular feature worthy of emulation across the industry (Amul Sustainability Report 2023-24, p. 53).



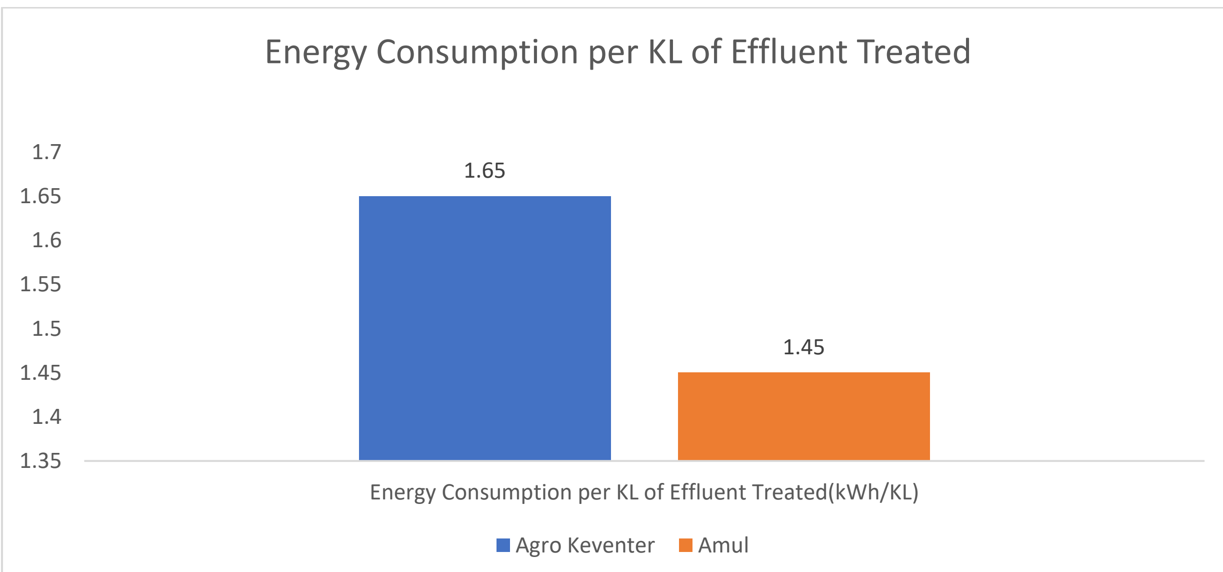


The above two graphs illustrate the comparative wastewater treatment infrastructure of Agro Keventer and Amul, specifically showing the differences in designed effluent treatment capacity and estimates of annual volume treated at both companies. These visualizations help highlight how each company's ETP scale aligns with their operational needs and sustainability commitments (Keventer Agro Limited, 2019; Amul, 2024).

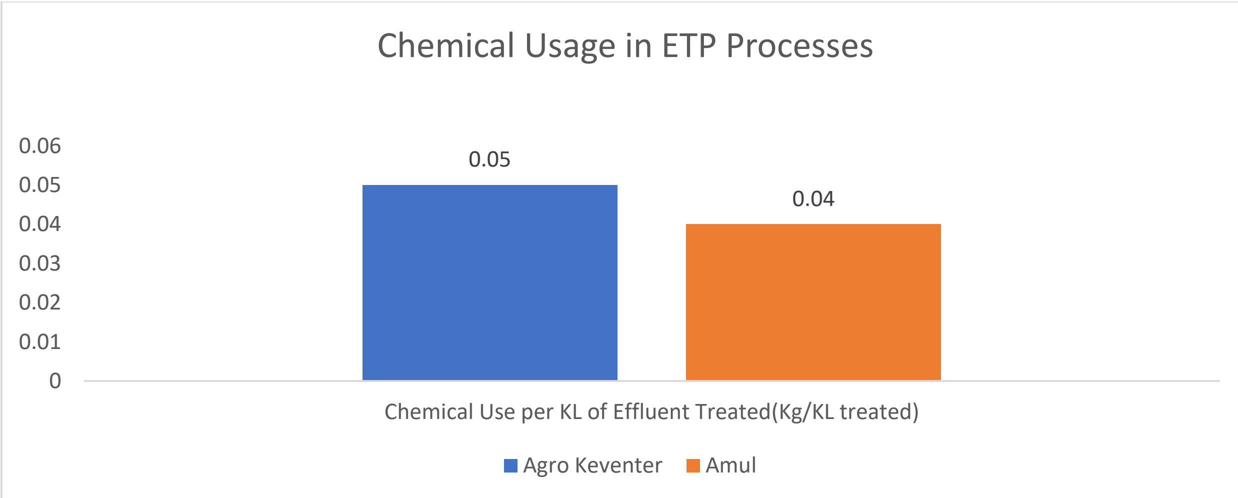
The treatment sequence at both companies begins with preliminary screening and equalization, followed by pH adjustment using chemicals such as lime and polyaluminum chloride (PAC). This is followed by coagulation and flocculation, where fine particles are aggregated and removed in clarifiers. The clarified effluent then undergoes biological treatment, primarily through aerobic processes. Both companies employ advanced technologies such as Moving Bed Biofilm Reactors (MBBR) to enhance the efficiency and resilience of biological treatment, especially under variable loading conditions (Ecologix Systems, 2025; PubMed, 2001).

Operational Efficiency

The efficiency in ETP operations can be gauged by parameters like energy use, chemical consumption, percentage recycling of water, and process upturn. Keventer's ETP utilizes around 1.65 kWh energy per kiloliter treated effluent, compared to slightly more efficient Amul plants consuming around 1.45 kWh per kiloliter, presumably due to larger scale and continuous process improvements (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24). Chemical application is also minimized with Keventer consuming around 0.05 kg chemicals per kiloliter compared to Amul's lower usage level of 0.04 kg per kiloliter indicative of constant efforts towards minimizing costs as well as reducing environmental footprint.

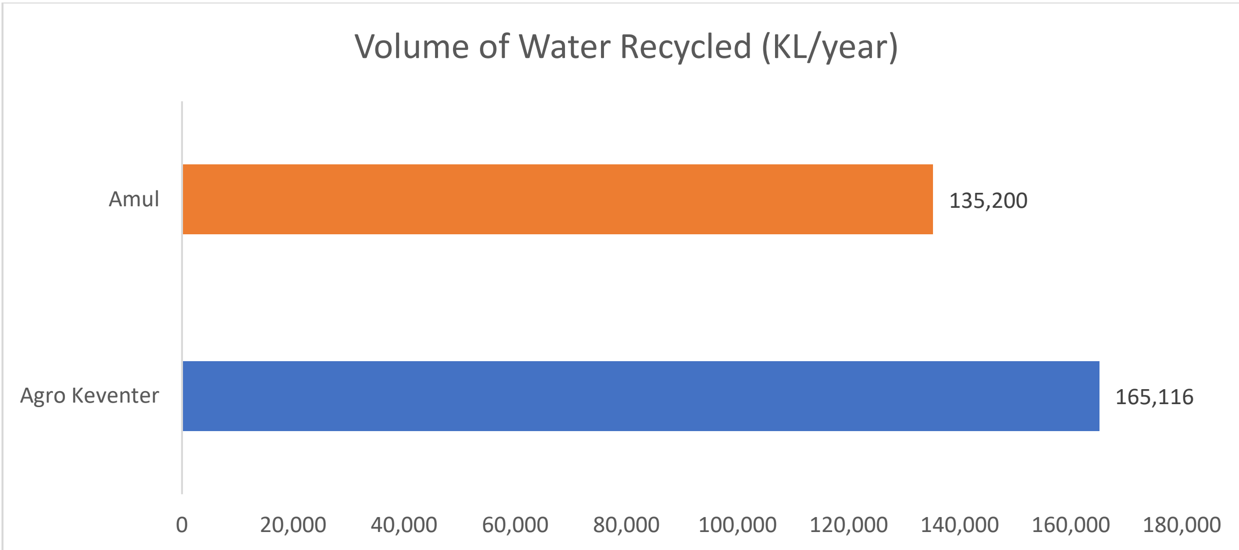


The above graph shows the energy consumed per kiloliter of effluent treated by ETPs at Agro Keventer and Amul, enabling a direct comparison of operational efficiency in power usage. Source: Keventer Agro Limited (2019); Amul (2024).



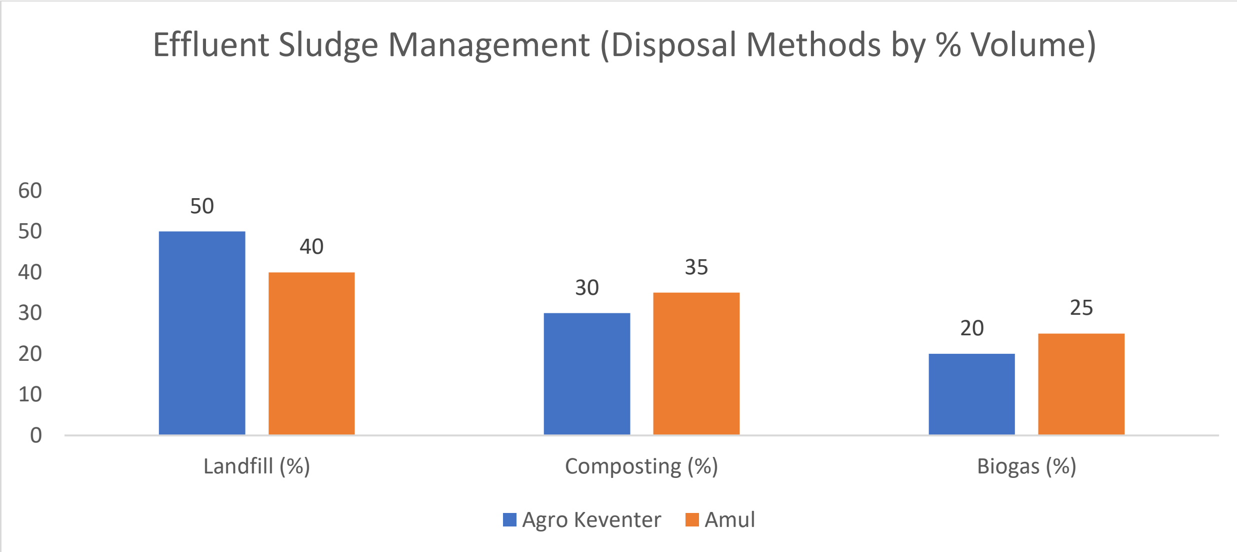
This graph compares the amount of chemicals used per kiloliter of effluent treated by Agro Keventer and Amul, providing insight into the efficiency of their ETP processes. Source: Keventer Agro Limited (2019); Amul (2024).

Water recycling is an important measure of operating effectiveness as well as sustainability. In 2018, Keventer reused more than 165,000 kiloliters of water for non-potable purposes like cooling, washing, and gardening, helping reduce its dependence considerably on groundwater (Keventer Sustainability Report 2019). In 2022, Amul reused 135,200 kiloliters, a number that reflects its diversification-oriented business model and credo towards water conservation (Amul Sustainability Report 2023-24). These companies have put in place tracking as well as optimizing systems for water use that incorporate sensors with real-time tracking as well as automatic controls that help in working more efficiently.



This graph shows the annual volume of water recycled by Agro Keventer and Amul, allowing a direct comparison of their efforts in resource conservation and water reuse. Source: Keventer Agro Limited (2019); Amul (2024).

Yet another confluence of sustainability and operating effectiveness is sludge management. Keventer accomplishes its sludge management by sending 50% to landfill, 30% to composting, and 20% for biogas production, offsetting waste disposal against resource recovery (Keventer Sustainability Report 2019). Amul has been aggressive in investing in biogas plants, employing anaerobic digestion to treat high-organic sludge as a source of renewable energy. Biogas in some Amul plants fulfills up to 5% total thermal energy requirements, illustrating a harmonious integration of waste-to-energy systems (Amul Sustainability Report 2023-24; Times of India, 2016).



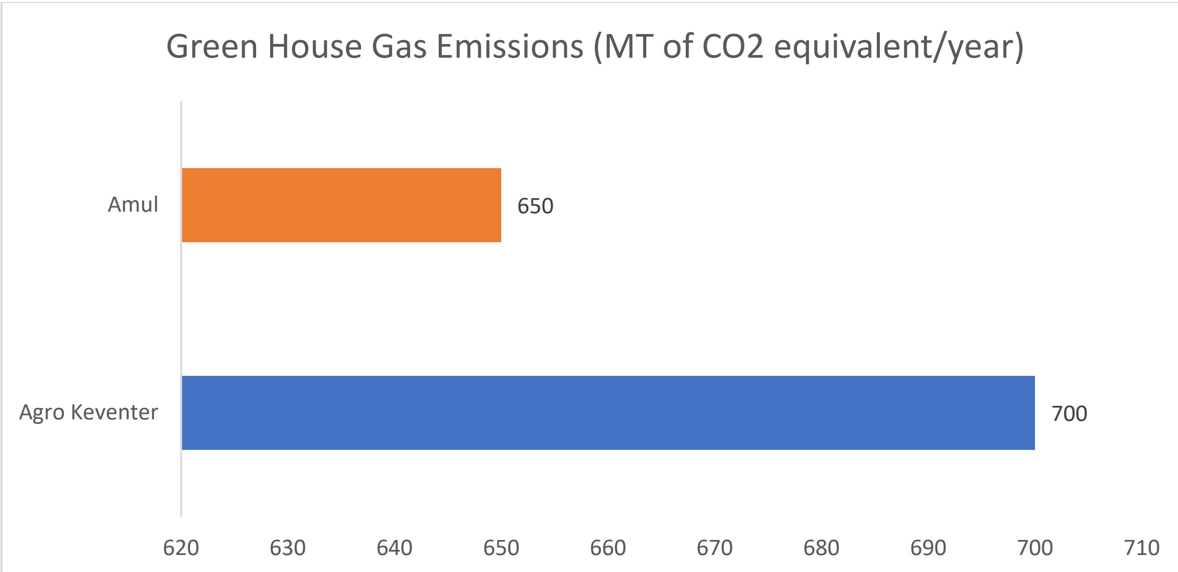
This graph compares the methods used by Agro Keventer and Amul to manage effluent sludge, showing the proportion directed to landfill, composting, and biogas generation. Source: Keventer Agro Limited (2019); Amul (2024).

Process uptime and maintenance practices also differ. Keventer schedules quarterly maintenance, resulting in about 30 hours of downtime per year, while Amul's monthly maintenance regime keeps downtime as low as 22 hours per year (Amul Sustainability Report 2023-24). The use of automation and advanced sensors at both companies supports high process reliability and minimizes manual intervention.

Sustainability Outcomes

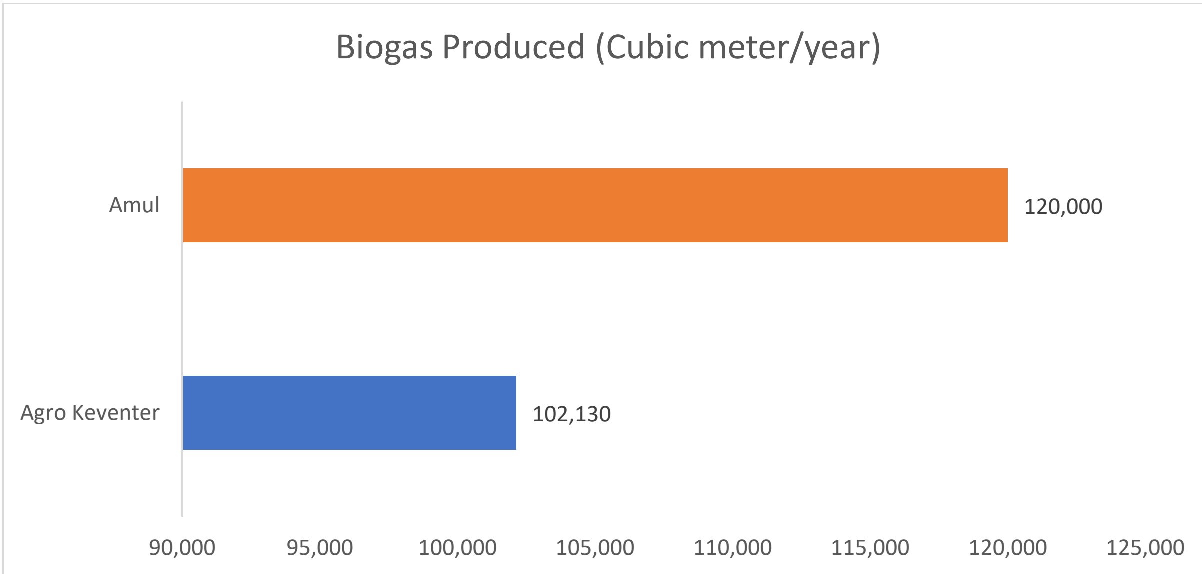
Sustainability indicators compare sustainability outcomes with such parameters as GHG emissions, resources recovered, and value addition to the circular economy. Keventer's ETP initiatives create up to 700 metric tons per year of CO_2 equivalent emissions, while Amul's bigger but more efficient ETP produces up to 650 metric tons per year (Keventer Sustainability

Report 2019; Amul Sustainability Report 2023-24). Both firms constantly make efforts towards bringing this down by investing in renewable energy as well as in enhancing process efficiencies.



This graph compares greenhouse gas emissions from ETP operations at Agro Keventer and Amul, highlighting their respective annual carbon footprints. Source: Keventer Agro Limited (2019); Amul (2024).

Both companies' sustainability approaches feature resource recovery. Keventer's treated water recycling and sludge composting help regenerate local ecosystems as well as augment agricultural activity. Amul produces 120,000 cubic meters of biogas per year through its ETP projects, while Keventer generates 102,130 cubic meters annually. Biogas projects by Amul reduce garbage in addition to offering a clean energy supply that reinforces its global sustainability agenda. The application of composted sludge as fertiliser by farm activities further completes the resource cycle as a demonstration of circular economy principles (DSF-IFAD, 2023; Keventer Agro Limited, 2019; Amul, 2024).



This graph presents a comparison of annual biogas production from ETP sludge at Agro Keventer and Amul, reflecting their efforts in energy recovery and waste-to-energy conversion. Source: Keventer Agro Limited (2019); Amul (2024).

Both firms also maintain robust reporting and monitoring systems for the environment. Keventer's value creation from sustainability has been estimated as around INR 373 crores annually, an aggregate value that incorporates the gains across water, energy, as well as waste reduction (Keventer Sustainability Report 2019). Though no separate amount for its sustainability efforts is declared by Amul, sheer size and technological prowess on its part indicate a substantial effect.

Comparative Insights

While both Agro Keventer and Amul have adopted advanced ETP technologies and best practices, there are notable differences in their approaches and outcomes. Amul's larger scale allows for greater investment in automation, process optimization, and resource recovery, resulting in slightly higher operational efficiency and sustainability performance. Its commitment to transparency, external audits, and continuous improvement sets a high standard for the industry.

Keventer, on a smaller scale, shows excellent water recycling and sustainability value creation that can be quantified. Its balanced sludge management approach as well as investment in state-of-the-art biological treatments make it a champion mid-sized dairy processor. Both processors struggle with further energy as well as chemical reductions, increasing biogas generation, as well as wider water reuse offsite.

The greatest challenge common to both firms is the imperative towards perpetual process improvement in terms of efficiency, resource recovery, as well as openness. Next-generation sector standards need to gravitate towards normalized recycle rates for wastewater, product-unit emissions rates, as well as integration of circular economy principles in every facet of ETP management (IRJET, 2019; DSF-IFAD, 2023).

Overall, the comparison between Agro Keventer and Amul's ETP operations shows that both firms value environmental compliance, operating efficiency, and sustainability. In terms of scale and spending on state-of-the-art technologies, Amul enjoys a small edge in operating efficiency and resource recovery; Keventer's emphasis on water recycle and sustainability value generation shows superior performance in its market category. Both firms set examples for the Indian milk industry on the value of sound ETP management in meeting environmental as well as business goals. Investment in further advances in technology, process improvements, and transparency will be required to sustain as well as improve their leadership in eco-friendly wastewater management.

7. Conclusion

The comparative analysis of effluent treatment plant (ETP) strategies in Agro Keventer and Amul signifies the significance of effective wastewater management in the Indian milk industry. Both organizations have been clearly diligent in terms of environmental compliance, efficiency in operation, as well as sustainability, and have established standards for the milk industry by embracing the use of upmarket technologies and best approaches (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24).

Agro Keventer's ETP features a multi-stage process involving physical, chemical, and biological treatment steps. The firm has invested in real-time monitoring, advanced biological reactors like Moving Bed Biofilm Reactors (MBBR), as well as automation that supports consistent adherence to regulatory standards. Keventer's emphasis on balanced sludge management, energy optimisation, and water recycling demonstrates a holistic approach towards conservation of resources as well as environment care. The firm's sustainability value creation in measurable terms and its periodic internal as well as third- party audits further strengthen its leadership story in the mid-sized dairy segment.

Operating on a much larger scale, Amul has used its resources to establish very efficient ETPs in its processing divisions. The application by the company of modern technologies like membrane filtration and massive biogas plants has allowed it to maintain enhanced process efficiency as well as recovery of resources. The reporting transparency by Amul, external certification, as well as constant process improvement, has helped it maintain over 99% compliance rates on a consistent basis. Investment by the company in biogas generation as well as sludge composting aids its shift towards a circular economy business approach, while its aggressive maintenance regime ensures minimal downtimes as well as robust process reliability.

In spite of these successes, both corporations struggle with further minimizing energy and chemical use, increasing water recycle utilization, as well as further integrating circular economy principles. Ongoing improvement in process effectiveness, resource recovery, and transparency becomes a necessity as standards for regulation increase in strength as well as expectations by stakeholders grow (IRJMETS, 2025; Awetech Works, 2024).

Based on these results, some recommendations can be made for Agro Keventer, Amul, and the overall dairy sector. First, firms need to keep investing in advanced treatment technologies like membrane filtration and anaerobic digestion in order to improve resource recovery as well as lower impacts on the environment. Second, the use of real-time monitoring and automation needs to increase as a means of maximizing process control and consistent conformance. Third, more attention needs to go toward water reuse offsite as well as in partnerships with immediate rural/agriculture stakeholders. Fourth, firms need to get more transparent as well as detailed in their sustainability reporting so that more precise benchmarking as well as more-informed decision-making can be ensured.

Drawing a conclusion, Agro Keventer's ETP and Amul's ETP can be examples for the Indian milk industry on how an environmentally responsible approach can be coupled with effective operation as well as sustainability by investing strategically, innovating, as well as improving continuously. Adopting best practices as well as meeting emerging challenges can help the milk processors create a more sustainable as well as a more resilient food system for India's future (Keventer Sustainability Report 2019; Amul Sustainability Report 2023-24; Awetech Works, 2024).

8. References

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